Cache Me if You Can: Repairing the KV State after Compression

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Abstract

As large language models (LLMs) support longer contexts and more capable agentic sessions, what matters in the accumulated context can change with each new query, tool output, test result, or browser observation. At the same time, inference still depends on a limited active key-value (KV) cache, and many existing KV-cache compression methods decide which past tokens stay active from the current prefix. As a result, tokens that matter in later turns may already have been compressed away in earlier compression decisions. We study post-compression KV cache repair, a runtime mechanism that restores previously evicted tokens to the active cache. RepairKV uses the interval between turns to reevaluate which tokens are important from evicted KV rows stored in a slower memory tier, then promotes a small subset before decoding resumes. Controlled experiments show the same repair effect under changing relevance and different initial eviction rules, and a controlled real-repository diagnostic shows the same pattern in a non-synthetic setting; on Qwen2.5-7B-Instruct at 32K context, RepairKV reaches 91.0% retrieval on a four-query needle-in-a-haystack task versus 24.5% for the matched no-repair baseline at the same active-cache budget, using only 96 promoted tokens.

1. Introduction

Motivation. Long-context inference increasingly depends on KV-cache compression and offload. During decoding, a transformer stores key-value (KV) activations for prior tokens so later tokens can attend to them. The model weights are fixed once loaded, but the KV cache grows with the number of tokens kept for each live sequence; at long context lengths, this sequence-dependent state can dominate available GPU memory and increase inference-time attention and cache-management costs. KV-cache compression and offload reduce this pressure by deciding which historical state remains active on GPU and which state is evicted or moved to a slower tier. That decision is usually made with information available at the current turn, so it is often treated as a one-way operation: evict now, decode later from the retained active set.

This static view misses two facts about multi-turn inference. First, attention relevance is query- and context-dependent: later turns, retrieved documents, tool outputs, test failures, or file changes can make previously low-priority context newly important. Query-aware and fine-grained KV retrieval systems show that both token criticality and attention-distribution patterns can change with the active query (Tang et al., 2024; Wang et al., 2025). Second, agentic workflows often contain non-decoding intervals while the model waits for external results. Recent systems make this operationally visible: Continuum schedules multi-turn agent requests around tool-call gaps and KV-cache lifetimes, while AgentServe targets local agent serving under tight GPU memory budgets (Li et al., 2026; Zhang et al., 2026). These systems show that turn boundaries, cache lifetime, and tool-call waits are now explicit systems objects, but they preserve or schedule cached state rather than repairing an already-compressed active cache using the next-turn signal.

Coding-agent measurements reinforce why this interval is worth studying: OS-level execution, including tool calls and setup, can account for 56–74% of task latency (Zheng et al., 2026). Together, changing relevance and non-decoding slack create the opportunity we study: use the pause before the next decode to revise active KV state after compression.

This paper studies whether that pause can improve the state seen by the next decode. In this setting, compression is not only a storage-saving step. It also decides which past tokens remain immediately available to the next decode. Once a request has paused, the runtime may hold a compressed active cache plus a host-memory warm buffer of evicted KV rows. If the next-turn cue becomes available before generation resumes, the system has an opportunity to revise which prior context is active. In shared serving this is a scheduler tradeoff; in the single-session or dedicated setting we target, the same interval is often idle time for the paused request.

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Either way, the central question is the same: can newly revealed turn-$(N{+}1)$ information repair a compressed cache under the same resumed active KV budget.

Setting. Recent analysis shows that KV compression can sharply degrade instructions and facts that later become important (Chen et al., 2025). The missing evaluation question is not simply whether compression loses information, or whether a serving system can preserve state across a pause. It is whether future-turn information can improve an already-compressed cache under the same resumed active-cache budget. A useful benchmark for this question needs explicit turn boundaries, a hidden future query, shared pre-turn history, and no-repair controls matched on resumed-cache budget $B_{\text{base}} + K$. In our experiments, the future relevance signal is an explicit second-turn (Q_2) retrieval query so that the repair effect is measurable, while the intended workflow claim is broader than query answering.

Contributions. REPAIRKV is an inference-time adaptation operator for tiered KV state, conditioned on the newly available next-turn signal. The adaptation is non-parametric: model weights, prompts, and decoding settings are fixed, while the active cache state is revised under a resource budget. The operator adds a reverse direction to the usual compression lifecycle: KV can be evicted or offloaded when memory is scarce, then selectively promoted when the relevance signal changes. After turn N, the runtime compresses the evictable context portion to an active set C_{base} with $|C_{\text{base}}| = B_{\text{base}}$ and retains evicted KV in a host-memory warm store W_N. When the turn-$(N{+}1)$ signal arrives, REPAIRKV scores that store and promotes a subset $S_K(Q_2) \subseteq W_N$, $|S_K| \leq K$, back into the active GPU cache before decoding resumes. This paper uses an explicit Q_2 retrieval query as the scoring signal because it gives a controlled first test of whether post-compression repair can exploit new relevance information. Unlike pause/resume serving, REPAIRKV changes the resumed active cache. Unlike within-turn query-aware compression, it starts from a paused compressed state and does not rerun full-prefix selection after Q_2 is revealed. We contribute:

- a matched resumed active-cache-budget protocol for the post-compression, pre-resume lifecycle stage, isolating cache repair from simply keeping more active tokens.
- calibrated evidence on pooled split-query multi-query needle-in-a-haystack (MQ-NIAH) 4Q/6Q/8Q evaluations, a five-turn revisit diagnostic, three first-stage retention rules, a Scissorhands-style persistence stress test, and preliminary real-repository diagnostic evidence, where REPAIRKV exceeds matched no-repair and content-agnostic controls while the real-repo variants remain below a label-assisted locality reference.
- a repair prototype with transfer/reinjection feasibility results and an instrumented latency probe that separates state movement, proxy scoring, and the diagnostic exact Q_2 attention-query scorer.

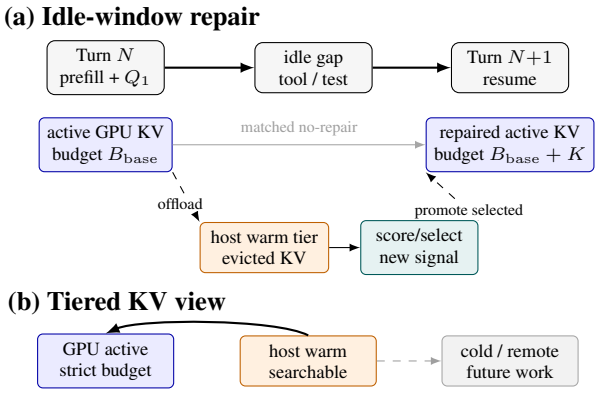

Figure 1. Systems-facing view of REPAIRKV as a paused-request KV-state adaptation primitive. A paused request offloads evicted KV to a host-memory warm tier, scores and promotes a budgeted subset after the next-turn signal arrives, and exposes a conceptual tiered-memory interface that production engines would materialize through page or block allocators.

This controlled scope identifies the repair mechanism on one primary model and one synthetic task family, with targeted retention-rule and real-repository checks. The adaptation-facing implication is that long-context agents may need between-turn KV state updates in addition to pre-generation pruning. Figure 1 gives the systems-facing data-path abstraction.

2. Related Work

Recent KV-cache compression and retrieval methods such as SnapKV, StreamingLLM, H_2O, and QUEST define the current frontier for memory-efficient long-context inference. SnapKV, StreamingLLM, and H_2O decide which past tokens remain active under a fixed budget, while query-aware methods such as QUEST emphasize that token criticality depends on the active query (Li et al., 2024a; Xiao et al., 2024; Zhang et al., 2023; Tang et al., 2024). Long-context and KV-lifecycle benchmarks stress contexts from ordinary long documents to 100K-token regimes, showing that application-visible context length does not remove the need for memory-management mechanisms (Bai et al., 2024; Zhang et al., 2024; Hsieh et al., 2024; Li et al., 2025). SCBench makes the KV-cache lifecycle itself a benchmark target, and ShadowKV shows that sparse KV reconstruction and offload can improve long-context serving throughput (Li et al., 2025; Sun et al., 2025). Fine-grained KV retrieval systems dynamically fetch query-relevant rows from larger stores during active inference (Tang et al., 2024; Wang et al., 2025; Qi et al., 2026); attention analyses likewise show that token importance can be query-dependent or recurrent rather than

fixed once compressed (Wu et al., 2024; Zhang et al., 2025). These works motivate dynamic KV state, but they mostly make the cache dynamic in one direction: retain less, offload more, or retrieve rows during active inference. REPAIRKV targets the complementary direction. It starts from an already compressed paused state, does not rerun full-prefix selection after Q_2 is revealed, and revises the active cache before decoding resumes under a matched resumed-cache budget.

Serving systems provide the substrate for this question. PagedAttention/vLLM improves GPU KV allocation and block management during serving (Kwon et al., 2023), while InferCept preserves paused requests to avoid recomputing prefixes when augmented generation resumes (Abhyankar et al., 2024). Agent and tool-use benchmarks make external actions part of the workload, including API calls, web interaction, repository edits, and interactive tasks (Qin et al., 2024; Zhou et al., 2024; Jimenez et al., 2024; Liu et al., 2024). Agent-serving and KV-tiering systems reason about resume prefills, offloading, cache reuse, and tiered storage across GPU, host memory, storage, and network tiers (Li et al., 2026; Zhang et al., 2026; Liu et al., 2025; Dzikanyanga et al., 2026). These systems manage where KV state lives and how it is reused. REPAIRKV asks which offloaded cached units should re-enter the active cache when a later turn reveals a new relevance signal.

3. Method

Overview. Each compression step chooses the active KV set using the information available at that point. Before a repaired cache is used for decoding, any available signal about the upcoming computation can guide which offloaded KV rows should re-enter the active cache. We study a minimal controlled repair mechanism: promote selected offloaded KV while avoiding a full query-aware recompression pass over the entire prefix. Figure 1 summarizes the systems-facing data-path abstraction. At the pause boundary, the abstraction has three objects:

- C_{base}: active evictable KV, with $|C_{\text{base}}| = B_{\text{base}}$;
- W_N: offloaded evicted KV, hidden from decoding unless restored;
- s_{N+1}: the newly available turn signal, instantiated as Q_2.

The repair map chooses $S_K(s_{N+1}) \subseteq W_N$ and resumes from $C_{\text{repair}} = C_{\text{base}} \cup S_K$.

Two-turn protocol. The protocol proceeds in two turns. The model first processes a long context and answers Q_1 from the full cache. After that answer, the runtime compresses only the evictable context portion to a base budget B_{base}, while preserving the exact Q_1 prompt and generated-answer tail. During the subsequent tool-call idle window, the turn-$(N{+}1)$ query Q_2 becomes known. REPAIRKV scores offloaded evicted KV against the active post-Q_1 cache and Q_2 signal, restores a K-budgeted subset S_K, and answers Q_2 from C_{repair}. The data path has two movements: evicted KV is offloaded to a host-memory tier after Q_1, and selected KV is copied back before Q_2 decoding. Full-prefix compression after Q_2 is revealed is reserved for the recompute baseline family. This isolates the adaptation operator for one paused request on one GPU: a single host-memory tier, the full Q_2 signal known before resumed decoding, and a pre-materialized evicted-KV store. Scheduling, remote tiering, and retention contention are separate systems questions.

Repair operation. The repair step scores evicted context positions against the newly available turn-$(N{+}1)$ signal, packs the highest-scoring anchors into short local bursts, and promotes the selected KV rows in the prototype back into the active cache before decoding resumes. The prototype restores local bursts because the query often contains a key-bearing anchor while the answer depends on nearby value tokens. Here K denotes a restored-context-token budget; anchor count varies with burst packing. The exact scorer forms attention scores between RoPE-rotated Q_2 queries and the union of active and evicted keys; writing $a_i = \text{mean}_{\ell,h} \max_t q_{2,t}^{(\ell,h)} \cdot k_i^{(\ell,h)}$, positions are ranked by a_i, with the turn-N score and absolute position used only as deterministic tie-breakers. Each anchor expands to a fixed local burst: two tokens left and twenty right. Duplicates are skipped, and the ranked list backfills until K unique evicted context rows are selected. The current prototype scores token rows, but the systems abstraction is promotion over cached units. A systems implementation would batch these promotions into exposed blocks or pages.

Matched-budget evaluation. All main claims are made against a no-repair baseline with the same resumed active-cache budget. A matched resumed-cache budget means that Q_2 decoding sees the same number of active evictable-context tokens in addition to the identical preserved Q_1 prompt and answer tail. It does not mean that the runtime stores the same amount of offloaded KV state. Let B_{match} denote the no-repair condition that applies the base eviction policy with context budget $B_{\text{base}} + K$ while preserving the same Q_1 prompt and answer tail. REPAIRKV decodes from a $B_{\text{base}} + K$ active context-row budget by starting from B_{base} retained rows and promoting K rows from the host-memory warm store in this prototype. The extra store bytes, scorer time, and transfer time are reported outside the matched active-cache budget. The comparison therefore measures active-cache fairness; total service cost is reported separately. The main restore-budget frontier reports raw answer score so readers can compare absolute quality across task difficulty levels. For effect-size comparisons, we also

report score gain over matched no-repair,

$$\Delta_{\text{repair}} = \text{Score}(\text{RepairKV}) - \text{Score}(B_{\text{match}}) \quad (1)$$

where Score is the mean fraction of held-out Q_2 targets recovered correctly per example, averaged over the fixed evaluation set shared across all conditions for that setting. This retrieval score measures target recall, while unconstrained answer quality is outside the controlled benchmark. Stricter exact-set and precision-sensitive scoring serve as confirmatory checks. For diagnostic purposes only, SpanRef-K uses benchmark metadata: it enumerates annotated Q_2 answer-span groups and selects the highest-scoring subset whose token cost is at most K. SpanRef-K is exact over this small span-group family. Broader K-token cache-state optima and upper bounds remain outside this diagnostic. Thus the protocol tests whether newly revealed turn-$(N{+}1)$ information can improve a compressed cache relative to matched no-repair retention under the same active-token budget.

4. Experimental Setup

Benchmark family. The main evaluation uses Qwen2.5-7B-Instruct (Qwen, 2024) at 32K rendered context on a single RTX PRO 6000 Blackwell GPU, with context-only SnapKV-style retention as the initial post-Q_1 compressor applied before repair (Li et al., 2024a). The task family is a split-query version of multi-query needle-in-a-haystack (MQ-NIAH) derived from RULER (Hsieh et al., 2024). Each needle binds a key to a value inside a long distractor context, and the model must return the values for the keys requested in the current turn. The 2Q, 4Q, 6Q, and 8Q variants use the same underlying task template. The name denotes how many key/value needles are queried across the two turns. Larger variants ask for more values and use longer decode budgets, but do not change the benchmark family. This synthetic family is designed for control: the future-relevant spans are annotated, the second-turn relevance shift is explicit, and every condition shares the same turn-N history. For each variant, we pool balanced, non-recency-favored partitions whose Q_2 side excludes the tail-recent needles. Appendix Figure 9 summarizes the exact splits.

Evaluation configuration. The main study avoids both recency saturation and the all-zero baseline regime found at smaller budgets. Unless stated otherwise, the fixed sweep is:

- compressor: context-only SnapKV with $R_{\text{ctx}} = 128$;
- base budgets: $B_{\text{base}} = 8192$ for 2Q, $B_{\text{base}} = 16384$ for 4Q, and $B_{\text{base}} = 18432$ for 6Q/8Q;
- restore budgets: $K = 8, 16, 24, 32, 48, 64, 80, 96, 128$;
- replay: one shared full-cache-generated Q_1 transcript per setting and condition.

This removes turn-N error-cascade differences so that the frontier isolates resumed-cache selection quality at turn $N{+}1$. MQ-NIAH-2Q/4Q/6Q use fixed 100-example evaluation sets per split; MQ-NIAH-8Q uses 24 examples per split over five partitions. Prompt and decode settings are reported in the appendix.

Scorers. The turn-N eviction signal is the exact generated Q_1 answer tail instead of a generic last-32-token cache suffix. The main curves use an exact Q_2 attention-query scorer that extracts the model's Q_2 self-attention query vectors after RoPE and scores them against active and evicted keys. The latency probe also measures a cheaper Q_2-appended-state proxy scorer, reported separately from the exact quality curves.

Baselines and references. Every condition reuses the same full-cache-generated turn-N transcript when answering Q_2. We compare the full-cache reference A, the base compressed cache B, matched no-repair B_{match}, Random-K and Oldest-K restore controls, and RepairKV. Random-K and Oldest-K restore from the same evicted-KV store as RepairKV without future-query scoring, so they test whether gains come from query-aware repair or generic reinsertion. The specificity study additionally uses StaleQ-K (score with the previous-turn query), WrongQ-K (score with another example's Q_2), and Refresh-buffered, a stronger Q_2-time reference that reselects the whole $B_{\text{base}} + K$ context budget from active plus offloaded rows without full-prefix recomputation. We use Refresh-buffered to expose selector headroom. The main claim is that idle-window repair is useful under matched active cache, not that the current selector is optimal among all Q_2-aware reselectors.

5. Results

Matched-budget frontier. Across the 2Q/4Q/6Q/8Q MQ-NIAH family, RepairKV separates from matched no-repair at moderate-to-large restore budgets under the same resumed active KV budget. Figure 2 reports the raw Q_2 answer-score frontier. Random-K and Oldest-K stay near matched no-repair in the same evaluations. MQ-NIAH-2Q saturates by $K = 80$, while matched no-repair remains near 0.06. The higher query-count settings separate the curves more gradually: at $K = 96$, RepairKV scores 0.910 on 4Q, 0.989 on 6Q, and 0.960 on 8Q, while matched no-repair scores 0.245, 0.422, and 0.546, respectively. This shows that generic reinsertion from the same offloaded evicted-KV store does not explain the effect. Appendix diagnostics report SpanRef-K headroom and operating regimes: on MQ-NIAH-6Q at $K = 48$, SpanRef-K scores 0.994 versus 0.653 for RepairKV.

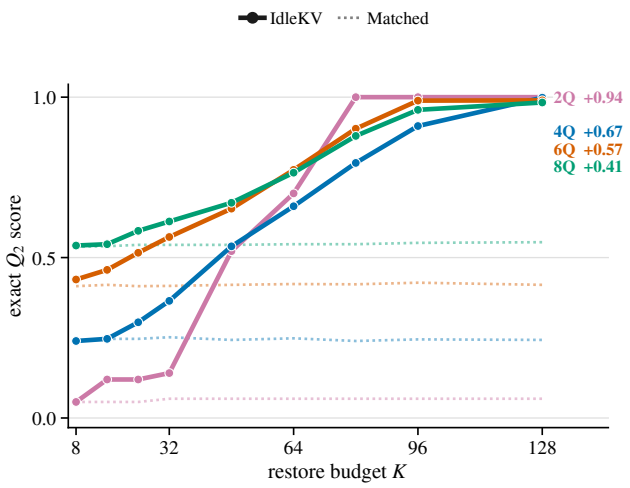

Figure 2. Raw exact Q_2 score under matched resumed-cache budgets. Solid curves are REPAIRKV. Dotted curves are matched no-repair. Right labels report query count and Δ_{96}. Figure 5 gives restored-span overlap. Appendix Figure 9 gives per-partition SpanRef-K endpoints.

Table 1. Next-turn signal specificity at $K = 48$ on MQ-NIAH-4Q (72 paired example-partition cases). Stale and donor controls test whether any query-like signal suffices; Refresh is a full-budget reselector over active plus offloaded KV.

Condition	Q_2 score	Δ vs. matched [95% CI]
Matched no-repair	0.215	–
StaleQ-K	0.243	+0.028 [−0.021, 0.083]
WrongQ-K	0.243	+0.028 [−0.014, 0.069]
REPAIRKV	0.542	+0.326 [0.243, 0.410]
Refresh-buffered	1.000	+0.785 [0.722, 0.847]

Next-turn signal specificity. Repair gains are specific to the newly revealed current-turn signal. At $K = 48$, StaleQ-K and WrongQ-K stay near matched no-repair, with gains 0.028 $[-0.021, 0.083]$ and $[-0.014, 0.069]$, while REPAIRKV reaches 0.326 $[0.243, 0.410]$. Refresh-buffered reaches raw score 1.000, showing that REPAIRKV is an incremental primitive with remaining selector headroom.

Repeated relevance shifts. The repair effect also survives repeated controlled relevance shifts. On a fixed five-turn MQ-NIAH-8Q revisit schedule ($K = 80$, $n = 24$), REPAIRKV gains 0.542 over matched no-repair on non-initial turns with paired bootstrap interval $[0.458, 0.620]$. Random-K and Oldest-K gain only 0.010 and 0.021. Stale-query repair remains below current-query repair, supporting repeated cue-conditioned adaptation under controlled relevance shifts. End-to-end task validation requires trace-based agent benchmarks.

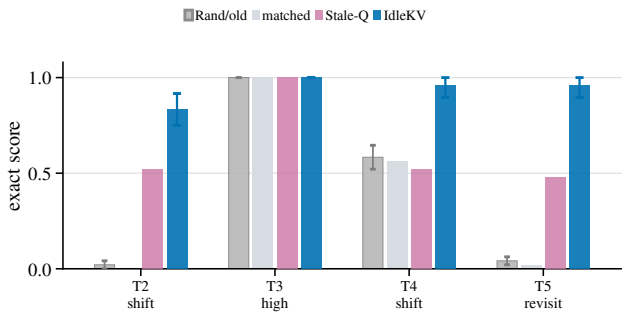

Figure 3. Five-turn relevance-shift diagnostic on MQ-NIAH-8Q (B=18,432, K=80, n=24). Groups show post-priming exact scores at equal active-cache budget. Intervals are bootstrap 95% CIs for REPAIRKV, and Stale-Q uses the previous-turn query.

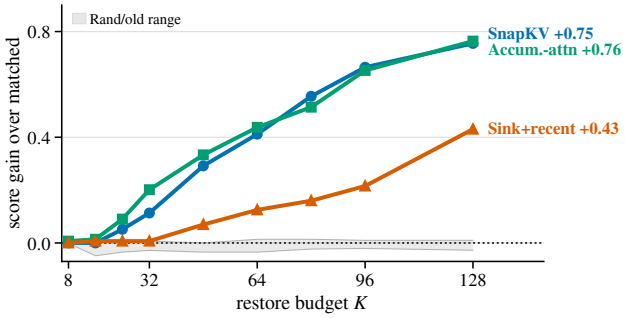

Figure 4. First-stage retention sensitivity on MQ-NIAH-4Q (n=24, B=16,384). Curves show exact Q_2 gain over each rule's matched no-repair baseline. The gray band spans Random-K/Oldest-K gains.

Retention-rule sensitivity. The adaptation effect persists under two protocol-matched retention heuristics beyond the primary SnapKV-style rule. The structurally different sink-plus-recent rule also passes the gate: at $K = 128$, REPAIRKV gains 0.431 over matched no-repair while Random-K/Oldest-K remain at the matched baseline. These are protocol-matched first-stage retention heuristics inspired by H_2O and StreamingLLM; canonical reproductions would require the full original online policies. A separate Scissorhands-style persistence-retention stress test (Liu et al., 2023) on MQ-NIAH-6Q also passes the locked gate (Appendix Table 3).

Robustness and mechanism. At $K = 128$, all twelve balanced partitions pass the direction check: REPAIRKV beats B_{match} and both content-agnostic restore controls, with minimum gains $0.585/0.387/0.292$ for 4Q/6Q/8Q. Appendix Figure 9 gives the per-partition endpoints. Figure 5 shows that REPAIRKV overlap with annotated Q_2 answer spans rises with K, while controls stay near baseline.

Real-repository diagnostic. To test external validity beyond MQ-NIAH, we ran a controlled real-repository

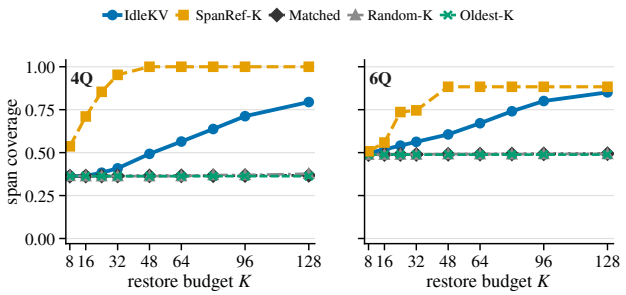

Figure 5. Mechanism diagnostic for the final exact MQ-NIAH evaluations. Curves show overlap between restored tokens and annotated Q_2 answer spans. Content-agnostic restores stay near baseline while REPAIRKV increasingly covers future-relevant spans.

relevance-shift diagnostic: 48 callsite examples from 12 pinned SWE-bench-pool repositories. Repair sees only an answer-redacted event cue, and all conditions decode the same event-plus-question prompt. At $K = 192$, event-only REPAIRKV improves exact identifier accuracy from 18.8% matched to 72.9%. Non-query controls and ToolFile-K remain near matched. File-gated REPAIRKV reaches 83.3%. Label-assisted AnchorWindow reaches 89.6%. This remains a small external-validity diagnostic, with no claim to be a real-code benchmark. Strict exclusions keep the gain positive ($+55.3/+62.9/+75.0$ points). Appendix Figure 7 gives the full audit.

Latency accounting. The exact Q_2 scorer is a mechanism diagnostic for the controlled quality curves, not the proposed runtime implementation. A Q_2-appended-state proxy provides benchmark evidence for a cheaper Q_2-conditioned scoring path: on paired 4Q/6Q runs at $K = 96$, it scores $0.970/0.894$ versus matched $0.245/0.422$. Appendix Figure 10 gives the proxy frontier. Production deployment would require GPU-side or paged warm-tier selection integrated with scheduler slack.

Table 2. Preliminary real-repository diagnostic at $K = 192$ over 48 callsite cases from 12 pinned SWE-bench-pool repositories. Accuracy is exact target-identifier recovery. Non-query summarizes the best of random, oldest, stale-cue, and wrong-event controls. AnchorWindow is label-assisted.

Condition	Accuracy
Matched no-repair	18.8%
Best non-query control	16.7%
ToolFile-K	18.8%
REPAIRKV	72.9%
File-gated REPAIRKV	83.3%
AnchorWindow-K	89.6%

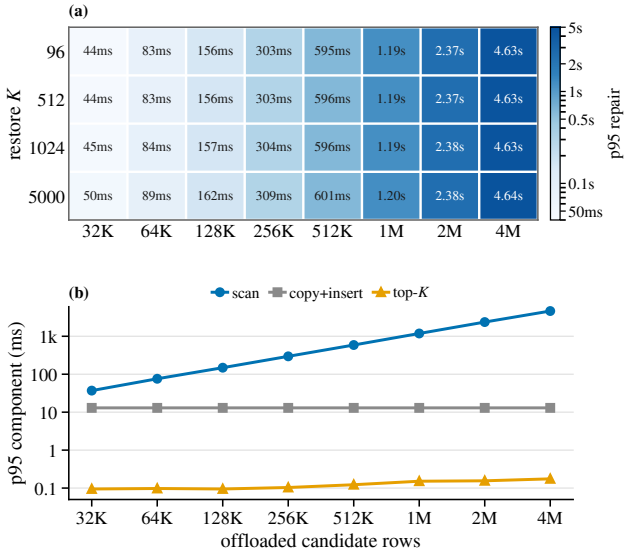

Figure 6. Runtime-capacity probe on the evaluation GPU. (a) Component-summed p95 service time at query length 64. (b) Stage-wise p95 component time for $K = 5000$. Generation and the research exact answer scorer are excluded.

Capacity envelope. Figure 6 reports a single-node p95 envelope for a chunked GPU selector over pinned host-memory BF16 keys. At $K = 5000$, p95 grows from 50.0 ms at 32K candidates to 1.20 s at 1M and 4.64 s at 4M. The plot should be read as a capacity envelope; trace-scheduled latency distributions remain a systems follow-up.

6. Discussion and Limitations

REPAIRKV is a between-turn maintenance primitive for making KV compression revisable: a runtime adaptation operator over active KV state for resource-adaptive tiered-KV runtimes. It complements existing compression, offload, query-aware access, and pause/resume serving systems (Kwon et al., 2023; Abhyankar et al., 2024; Liu et al., 2025; Dzikanyanga et al., 2026; Li et al., 2026; Zhang et al., 2026) by asking how active memory can be adapted after compression but before resumed decoding. The mechanism should be most useful when:

- the next turn changes which old context matters;
- the compressed cache still leaves useful choices in the evicted warm tier; and
- the idle window or scheduler has enough slack for scoring and KV movement.

The experiments isolate one paused request on one GPU, matching local or dedicated agent inference but omitting multi-tenant contention. In shared serving, a scheduler would still need to account for HBM blocks, copy bandwidth, and queueing slack. The quality comparisons match active GPU KV budget; total host-plus-device storage and

end-to-end service cost are reported separately. The exact scorer is a diagnostic rather than a production selector, the proxy scorer is only a first approximation, and stronger Q_2-aware reselectors or full-prefix recompute may dominate when enough idle time is available. The real-repository diagnostic is small and exploratory, so it supports external validity without serving as a SWE-bench issue-resolution benchmark. Next experiments should focus on trace-scheduled systems repair with page-level or learned selectors, plus stricter evaluation with time/compute-matched recompute baselines.

Conclusion. REPAIRKV makes the case for revisable KV compression. If context relevance changes after eviction, the cache lifecycle can move in both directions: compress to fit the active budget, then repair by promoting relevant evicted state before the next decode. Across MQ-NIAH-4Q/6Q/8Q, this repair beats matched no-repair while random and oldest-token restores stay near baseline. The broader direction is mutable KV memory for resource-adaptive long-context inference, with pause windows treated as part of the inference budget.

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A. Additional Evaluation Details

Protocol details. Concise split prompts request values only, comma-separated, with a short answer prefix and fixed decode budgets of 24 tokens for MQ-NIAH-4Q, 48 tokens for MQ-NIAH-6Q, and 64 tokens for MQ-NIAH-8Q. Generation is deterministic greedy decoding with no sampling. The main examples use dataset seed offset 0. Random-K uses a deterministic example- and K-dependent seed. Bootstrap intervals are percentile intervals over paired example-partition rows with fixed seeds. Main Qwen runs use BF16 weights on one RTX PRO 6000 Blackwell GPU with Python 3.12.3, PyTorch 2.10.0+cu128, and Transformers 5.2.0. The exact Q_2 attention-query scorer extracts the model's actual self-attention query vectors for Q_2 prompt tokens after RoPE and scores them against active and evicted keys. This is distinct from decode-time answer-token attention. The Q_2-appended-state proxy instead appends Q_2 to the compressed cache with an ordinary forward pass and reuses the appended Q_2 cache rows as scoring rows. Both scorers are benchmark instrumentation rather than fused production serving kernels.

Span-group diagnostic. $\mathrm{SpanRef}$-K starts from the same B state and offloaded evicted-KV store as REPAIRKV. It enumerates the small family of annotated Q_2 answer-span-group subsets, one key/value span group per requested needle, evaluates each subset once under actual Q_2 generation, and chooses the highest-scoring subset with total cost at most K. It is exact over this benchmark-specific span-group family, but not over all arbitrary token subsets or possible final generations. Because the selected span-group subset may use fewer than K rows, a repair selector can occasionally score higher at low budgets by spending the full budget on useful local neighborhoods around future-relevant spans. Thus $\mathrm{SpanRef}$-K is a diagnostic span-group reference, not a ceiling.

Partition choices. For MQ-NIAH-4Q, the pooled panel uses $14{\rightarrow}23$, $24{\rightarrow}13$, and $34{\rightarrow}12$. These are the balanced $2|2$ partitions whose Q_2 side excludes the tail-anchored fourth needle. For MQ-NIAH-6Q, the pooled panel uses $156{\rightarrow}234$, $256{\rightarrow}134$, $356{\rightarrow}124$, and $456{\rightarrow}123$, so the second turn excludes the two latest needles. For MQ-NIAH-8Q, the pooled panel uses $5678{\rightarrow}1234$, $1678{\rightarrow}2345$, $2678{\rightarrow}1345$, $3678{\rightarrow}1245$, and $4678{\rightarrow}1235$, again excluding the two latest needles from the second turn. Within-turn order is not treated as a separate condition because the prompt asks for comma-separated values only and scoring checks whether each requested value is recovered. Recency-favorable splits such as $12{\rightarrow}34$ are left to separate diagnostics outside the main future-query repair result.

Scissorhands-style retention stress test. Table 3 repeats the MQ-NIAH-6Q locked protocol after replacing the initial SnapKV-style retention rule with a one-shot persistence rule inspired by Scissorhands. The test asks whether repair still works when the compressed active cache is chosen by past attention persistence in place of the primary retention rule. Canonical online per-head Scissorhands reproduction remains future work.

Table 3. Scissorhands-style persistence-retention stress test on MQ-NIAH-6Q (B=18,432, n=24). Values are exact scores. Best ctrl. is the stronger of Random-K and Oldest-K; the interval is the paired bootstrap 95% CI for REPAIRKV$-B_{\mathrm{match}}$.

K	Full	B_{match}	Best ctrl.	REPAIRKV	Gain CI
48	0.990	0.365	0.382	0.712	[0.292, 0.403]
64	0.990	0.365	0.382	0.795	[0.378, 0.490]
80	0.990	0.378	0.385	0.934	[0.503, 0.604]
96	0.990	0.375	0.382	0.993	[0.566, 0.670]
128	0.990	0.368	0.378	0.990	[0.573, 0.674]

B. Additional Discussion

Tiered-cache scaling. Repair creates a two-level retention problem because the main curves match the resumed active GPU KV budget while tracking total host-plus-device storage separately. A strict device-resident policy chooses the active KV budget, while a looser off-device policy decides which evicted rows remain searchable, compressed, summarized, or recomputable. In the Qwen-shaped accounting above, 4M searchable rows already imply about 240 GB before metadata. Thus REPAIRKV supports the middle operation in a tiered hierarchy: turn-conditioned promotion from a warm tier during an idle gap. This is a hardware-facing implication; hardware measurement remains future work.

Idle-window repair interface. A practical implementation would expose a warm-tier index over evicted KV units, a time-budgeted selector conditioned on the next-turn signal, asynchronous promotion into the active cache, and fallbacks to

no-repair or full recompute. This turns a pause into a schedulable resource: the system can spend only the slack that remains after queueing, transfer, and generation constraints, and can move the selector from token rows to pages or compressed metadata as hardware and serving constraints require.

Future benchmark axes. Future benchmarks should vary the number, timing, and conflict structure of new-turn constraints. The 2Q and 8Q breadth checks suggest that easy regimes can saturate while stricter multi-query turns expose how quickly eviction failures accumulate. The five-turn diagnostic strengthens the dynamic-workflow evidence, but it remains synthetic. Broader task families should keep the same matched-budget controls while adding real tool traces and stronger selector families.

C. Supplementary Experimental Views

The supplementary figures probe external validity, query-count and operating-regime breadth, partition consistency, and latency/proxy behavior. Exact audit values remain in released result artifacts.

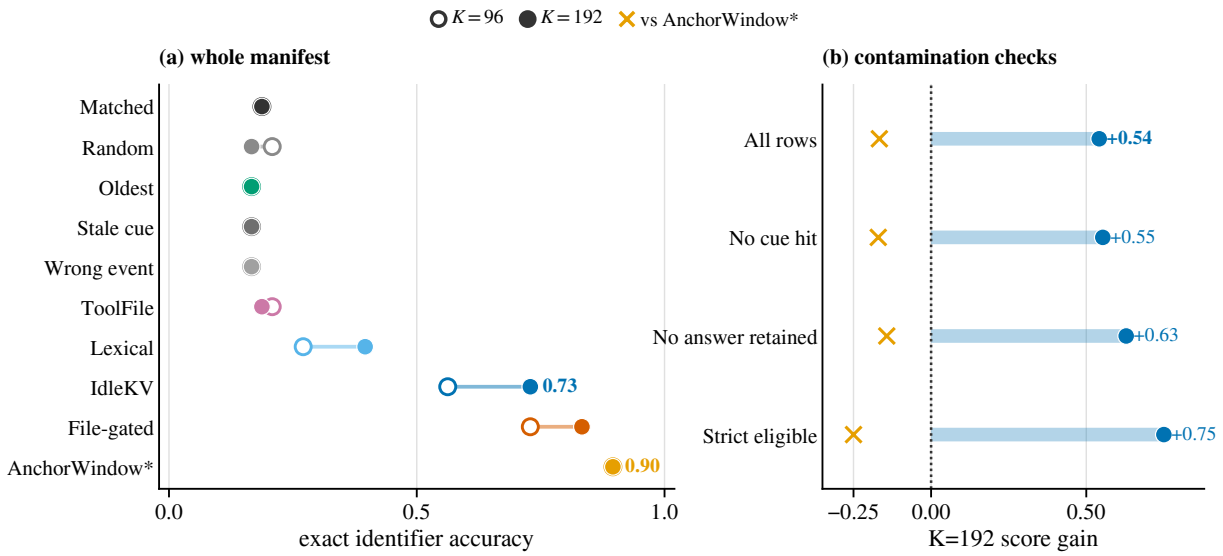
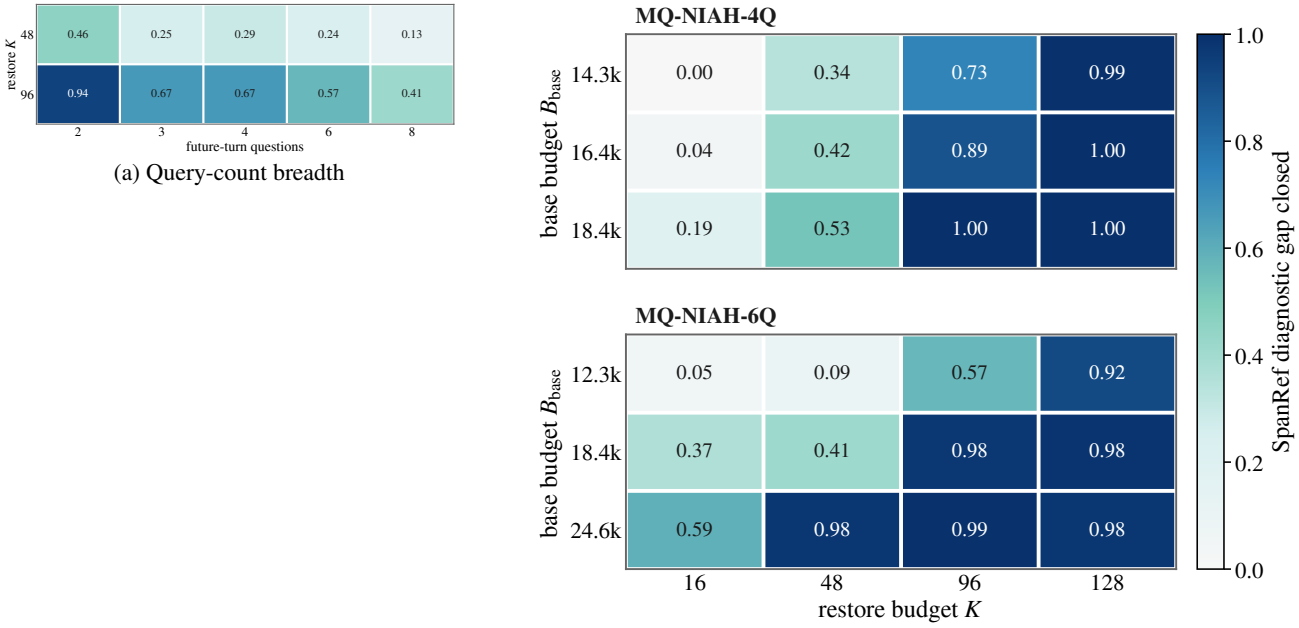


Figure 7. Controlled real-repository relevance-shift diagnostic over 48 callsite examples from 12 repositories drawn from the SWE-bench repository pool at pinned commits. Repair sees only the event cue. All conditions decode the same event-plus-question prompt. Left: exact identifier accuracy at $K = 96$ and $K = 192$. Right: $K = 192$ REPAIRKV gain over matched no-repair after removing cue-only or answer-retention rows. ToolFile-K is file-name-assisted with oldest-row backfill. File-gated REPAIRKV restricts candidates to the event-named file before applying the event-only scorer. Lexical is an answer-sanitized diagnostic outside the deployable-selector comparison. AnchorWindow-K is a label-assisted locality reference outside the deployable-baseline and end-to-end issue-resolution comparisons.



(a) Query-count breadth

(b) Operating regime

Figure 8. Supplementary breadth diagnostics. (a) Cells show REPAIRKV score gain over matched no-repair for $K = 48$ and $K = 96$ across 2-, 3-, 4-, 6-, and 8-question split-query variants. (b) Cells show the fraction of available SpanRef-K diagnostic gain recovered by REPAIRKV, clipped to $[0, 1]$, for MQ-NIAH-4Q/6Q. Saturated cells with little matched-to-SpanRef-K gap should not be read as upper-bound statements.

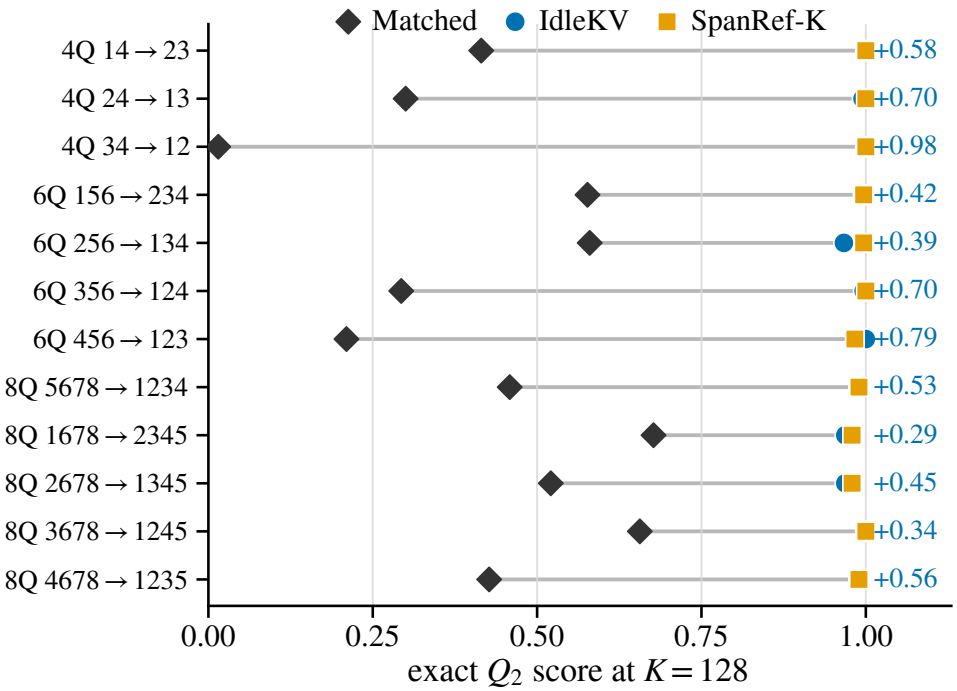
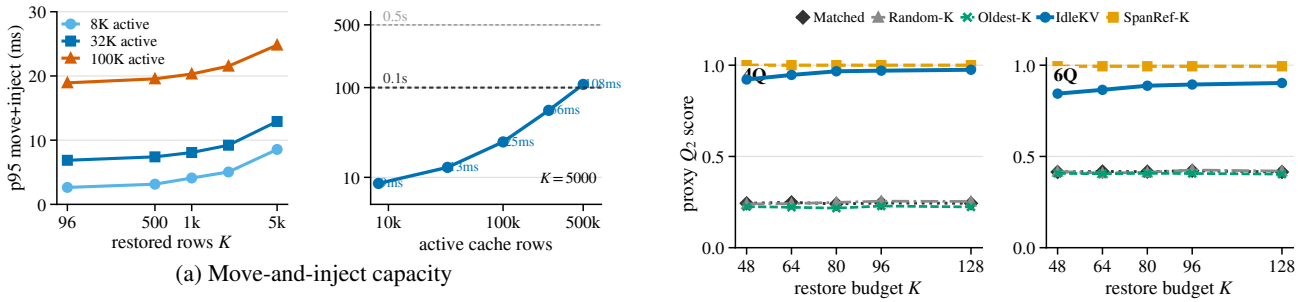


Figure 9. Per-partition endpoint scores at $K = 128$ on the final exact evaluations. Gray segments show the change from matched no-repair to REPAIRKV, orange points show the benchmark-metadata SpanRef-K diagnostic, and labels report the REPAIRKV difference over matched no-repair.



(a) Move-and-inject capacity

(b) Proxy scorer frontier

Figure 10. Supplementary latency and proxy diagnostics. (a) Synthetic move-and-inject capacity for Qwen-shaped BF16 KV on the evaluation GPU. Measurements use pinned host-memory restored blocks, exclude scorer time and generation, and report p95 latency as K or active cache size grows. (b) Controlled proxy-scorer frontier on MQ-NIAH-4Q/6Q (100 examples per split). At $K = 96$, content-agnostic restores stay within 0.010/0.004 of matched no-repair while proxy REPAIRKV remains well above matched.